

Verification of the t -graded Assaf–González Block-Triangular Structure of Nil-Hecke Atoms at Four Cases (μ, n)

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Abstract

The nil-Hecke atom polynomial $A_\gamma^{\text{alt}}(x_1, \dots, x_n; t)$ (defined via iterated θ_i^{alt} from x^μ) admits a conjectured block-triangular structure with respect to the Bruhat order on the $\mathfrak{S}_n$ -orbit of μ . This structure was verified at $\mu = (2, 2, 0)$, $n = 3$ during the 2026-07-24 wake and used to conjecture Route δ^+ — a t -graded structural closure of the $V_{<\mu}$ consistency lemma.

We verify the conjecture at three further cases: $\mu = (2, 1, 0)$, $n = 3$ (all distinct parts, $|\text{orbit}| = 6$); $\mu = (2, 2, 1)$, $n = 3$ (stabiliser $\langle s_1 \rangle$, $|\text{orbit}| = 3$); and $\mu = (2, 2, 0, 0)$, $n = 4$ (stabiliser $\langle s_1 \rangle$ in $\mathfrak{S}_4$, $|\text{orbit}| = 6$, $|B(\mu)| = 20$).

All four cases pass the *weak* form (support inclusion in the Bruhat-filtered subspace) and the *strong* form (diagonal block $(1, 1 - t, \dots, 1 - t)$; subdiagonal shadows divisible by t) wherever the diagonal/subdiagonal position is unambiguous. The global identity $\sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}} = P_\mu(x; t)$ is verified in all four cases, with $c_\gamma(t) \in \mathbb{Z}_{\geq 0}[t]$ following a Gaussian pattern.

A new structural feature emerges at $n = 4$: the multiplicity-2 weight $\nu = (1, 1, 1, 1)$ carries coefficient $(1 - t)^2$ rather than the naive $(1 - t)$. This *refines* the original conjecture: above-diagonal contributions in Bruhat are not identically zero at generic t ; instead, they are t -divisible. The $t = 0$ specialisation of the block structure remains exact, reproducing the classical Mason atom decomposition $B(\mu) = \bigsqcup_\gamma A_\gamma$.

1 Setup

Fix $n \geq 2$, indeterminate t , and variables $x_1, \dots, x_n$. Write s_i for the simple transposition $(i, i + 1)$ acting on positions.

Definition 1 (Nil-Hecke atom operator). For $1 \leq i \leq n - 1$,

$$\theta_i^{\text{alt}}(f) = \frac{x_{i+1} - t x_i}{x_i - x_{i+1}} \cdot (f - s_i(f)).$$

Since $f - s_i(f)$ is divisible by $x_i - x_{i+1}$, the operator sends polynomials to polynomials. At $t = 0$, $\theta_i^{\text{alt}} = x_{i+1} \cdot \partial_i$ is Mason’s classical atom operator [?].

Definition 2 (Nil-Hecke atom polynomial). Let $\mu \in \mathbb{N}^n$ and $\gamma \in \mathfrak{S}_n \cdot \mu$. Choose a Bruhat-shortest word $s_{i_l} \cdots s_{i_1}$ in $\mathfrak{S}_n$ taking μ to γ (bubble word $(i_1, \dots, i_l)$). Define

$$A_\gamma^{\text{alt}} = \theta_{i_l}^{\text{alt}} \cdots \theta_{i_1}^{\text{alt}}(x^\mu).$$

This is independent of the choice of Bruhat-shortest word because θ_i^{alt} satisfies the braid relations on words that stay reduced.

Definition 3 (Classical Demazure atom). For $\gamma \in \mathfrak{S}_n \cdot \mu$, the (right-key) atom $A_\gamma \subset B(\mu)$ is the subset of semistandard Young tableaux of shape $\text{sort}(\mu)$ characterised equivalently by

$$A_\gamma = B_{w_\gamma} \setminus \bigcup_{w' < w_\gamma} B_{w'},$$

where B_w denotes the Demazure sub-crystal at w , and w_γ the Bruhat-shortest element mapping μ to γ . The classical atoms partition $B(\mu)$. At $t = 0$, $A_\gamma^{\text{alt}}|_{t=0} = \sum_{T \in A_\gamma} x^{\text{wt} T}$ (L.-S. 1990 / Mason 2006).

Definition 4 (Bruhat-filtered subspace). For $\gamma \in \mathfrak{S}_n \cdot \mu$, set

$$F_\gamma := \mathbb{Q}(t)\text{-span}\{x^\nu : \nu \in \bigcup_{\gamma' \leq_B \gamma} \text{wt}(A_{\gamma'})\}.$$

2 The conjecture (weak, strong, and refined forms)

Conjecture 5 (Block-triangular structure, weak form). *For all $\gamma \in \mathfrak{S}_n \cdot \mu$, $A_\gamma^{\text{alt}} \in F_\gamma$.*

Conjecture 6 (Strong form, diagonal + subdiagonal shadows). *The coefficient of x^ν in A_γ^{alt} , denoted $[x^\nu]A_\gamma^{\text{alt}}$, satisfies:*

- If $\nu = \gamma$: $[x^\nu]A_\gamma^{\text{alt}} = 1$.
- If $\nu \in \text{wt}(A_\gamma)$, $\nu \neq \gamma$, and $\nu \notin \text{wt}(A_{\gamma'})$ for any $\gamma' \neq \gamma$: $[x^\nu]A_\gamma^{\text{alt}} = 1 - t$.
- If $\nu \in \text{wt}(A_{\gamma'})$ with $\gamma' <_B \gamma$ and $\nu \notin \text{wt}(A_{\gamma''})$ for any $\gamma'' \not\leq_B \gamma'$ or $\gamma'' \geq_B \gamma$: $t \mid [x^\nu]A_\gamma^{\text{alt}}$ in $\mathbb{Z}[t]$.

Conjecture 7 (Refined form: mod- t block-triangularity). *There exists a canonical decomposition $A_\gamma^{\text{alt}} = \sum_{T \in B(\mu)} c_{\gamma,T}(t) \cdot [T]$, where $[T]$ is a formal atom-label projecting to $x^{\text{wt}(T)}$, such that:*

- $c_{\gamma,T_\gamma}(t) = 1$ for the extremal element T_γ of A_γ ;
- $c_{\gamma,T}(t) = 1 - t + t \cdot (\text{correction})$ for $T \in A_\gamma \setminus \{T_\gamma\}$;
- $c_{\gamma,T}(t) \in t \cdot \mathbb{Z}[t]$ for $T \notin A_\gamma$ (*including both below and above in Bruhat, refining Conjecture ??*).

In particular, at $t = 0$, $c_{\gamma,T}(0) = \mathbf{1}_{T \in A_\gamma}$, recovering the classical Mason atom characteristic function.

Conjecture ?? weakens PROVE.md's original claim that above-diagonal contributions are 0, replacing it with “ t -divisible”. The multiplicity-2 weight case at $n = 4$ (Section ??) is what forces this refinement.

3 Verification data: four cases

We used a Python + SymPy implementation of θ_i^{alt} , the Kashiwara crystal $B(\mu)$, Bruhat order via Ehresmann's tableau criterion, and Demazure crystal enumeration. Code lives at `~/projects/probes/2026-07-24`

3.1 Case 0: $\mu = (2, 2, 0)$, $n = 3$ (sanity check)

This reproduces the verified 07-24 case. $|B(\mu)| = 6$, $|\text{orbit}| = 3$, atoms of sizes 1, 2, 3. Extract of the coefficient matrix $M(\mu)_{\gamma, \nu}$ (rows indexed by γ in Bruhat order, columns grouped by which atom contains ν):

	$A_{x^{(2,2,0)}}^{(2,2,0)}$	$A_{x^{(2,1,1)}}^{(2,0,2)}$	$A_{x^{(2,0,2)}}^{(2,0,2)}$	$x^{(1,2,1)}$	$A_{x^{(1,1,2)}}^{(0,2,2)}$	$x^{(0,2,2)}$
$A_{(2,2,0)}^{\text{alt}}$	1	0	0	0	0	0
$A_{(2,0,2)}^{\text{alt}}$	$-t$	$1-t$	1	0	0	0
$A_{(0,2,2)}^{\text{alt}}$	0	$t(t-1)$	$-t$	$1-t$	$1-t$	1

Block-lower-triangular in Bruhat on γ : diagonal is $(1, [1-t \ 1], [1-t \ 1-t \ 1])$, subdiagonals are pure t -divisible shadows, above-diagonal is 0. Global identity: $[3]_t A_{(2,2,0)}^{\text{alt}} + [2]_t A_{(2,0,2)}^{\text{alt}} + A_{(0,2,2)}^{\text{alt}} = m_{(2,2)} + (1-t) m_{(2,1,1)} = P_{(2,2)}(x; t)$.

3.2 Case 1: $\mu = (2, 1, 0)$, $n = 3$

All distinct parts. $|B(\mu)| = 8$, $|\text{orbit}| = 6$, Bruhat lengths 0, 1, 1, 2, 2, 3. Atoms:

$$\begin{aligned} A_{(2,1,0)} &= \{[[1, 1], [2]]\}, & A_{(2,0,1)} &= \{[[1, 1], [3]]\}, & A_{(1,2,0)} &= \{[[1, 2], [2]]\}, \\ A_{(0,2,1)} &= \{[[1, 2], [3]], [[2, 2], [3]]\}, & A_{(1,0,2)} &= \{[[1, 3], [2]], [[1, 3], [3]]\}, \\ A_{(0,1,2)} &= \{[[2, 3], [3]]\}. \end{aligned}$$

Notable structural feature: The weight $\nu = (1, 1, 1)$ appears in two SSYT: $[[1, 2], [3]] \in A_{(0,2,1)}$ and $[[1, 3], [2]] \in A_{(1,0,2)}$. These two atoms are *incomparable* in Bruhat, so $\nu = (1, 1, 1)$ lives in two Bruhat-incomparable atoms. The block-triangular structure handles this cleanly.

Explicit A_{γ}^{alt} (support and coefficients):

$$\begin{aligned} A_{(2,1,0)}^{\text{alt}} &= x^{(2,1,0)} \\ A_{(2,0,1)}^{\text{alt}} &= -t \cdot x^{(2,1,0)} + x^{(2,0,1)} \\ A_{(1,2,0)}^{\text{alt}} &= -t \cdot x^{(2,1,0)} + x^{(1,2,0)} \\ A_{(0,2,1)}^{\text{alt}} &= t^2 x^{(2,1,0)} - t x^{(2,0,1)} - t x^{(1,2,0)} + (1-t) x^{(1,1,1)} + x^{(0,2,1)} \\ A_{(1,0,2)}^{\text{alt}} &= t^2 x^{(2,1,0)} - t x^{(2,0,1)} - t x^{(1,2,0)} + (1-t) x^{(1,1,1)} + x^{(1,0,2)} \\ A_{(0,1,2)}^{\text{alt}} &= -t^2(t+1) x^{(2,1,0)} + t^2 x^{(2,0,1)} + t(t+1) x^{(1,2,0)} \\ &\quad + t(t-1) x^{(1,1,1)} + (-t) x^{(0,2,1)} + (-t) x^{(1,0,2)} + x^{(0,1,2)} \end{aligned}$$

The two Bruhat-length-1 rows $A_{(2,0,1)}^{\text{alt}}$ and $A_{(1,2,0)}^{\text{alt}}$ each have their own extremal (1) plus a single $-t$ subdiagonal shadow onto $A_{(2,1,0)}$. Consistent with strong block-triangularity.

The two Bruhat-length-2 rows each show the pattern: t^2 shadow on $A_{(2,1,0)}$, $-t$ shadows on $A_{(2,0,1)}, A_{(1,2,0)}$, diagonal block $(1-t, 1)$ on their own atom, and 0 on the incomparable-atom weights (*including* $x^{(1,1,1)}$ from the other A_{γ} containing $(1, 1, 1)$). The coefficient $(1-t)$ on $x^{(1,1,1)}$ in row $A_{(0,2,1)}^{\text{alt}}$ is entirely attributed to the SSYT $[[1, 2], [3]] \in A_{(0,2,1)}$ — the incomparable atom contribution is 0.

Global identity: $\sum c_{\gamma} A_{\gamma}^{\text{alt}} = P_{(2,1)}(x_1, x_2, x_3; t)$ verified with

$$c_{(2,1,0)} = [2]_t [3]_t, \quad c_{(2,0,1)} = [2]_t^2, \quad c_{(1,2,0)} = [3]_t, \quad c_{(0,2,1)} = [2]_t, \quad c_{(1,0,2)} = [2]_t, \quad c_{(0,1,2)} = 1.$$

Verdict: Conjectures ??, ??, ?? all PASS.

3.3 Case 2: $\mu = (2, 2, 1)$, $n = 3$ (stabiliser $\langle s_1 \rangle$)

Non-trivial stabiliser. $|B(\mu)| = 3$, $|\text{orbit}| = 3$, all atoms of size 1.

$$\begin{aligned} A_{(2,2,1)}^{\text{alt}} &= x^{(2,2,1)} \\ A_{(2,1,2)}^{\text{alt}} &= -t \cdot x^{(2,2,1)} + x^{(2,1,2)} \\ A_{(1,2,2)}^{\text{alt}} &= -t \cdot x^{(2,1,2)} + x^{(1,2,2)} \end{aligned}$$

Every diagonal block is 1×1 (just extremal); all subdiagonal shadows are $\pm t$ or 0.

Global identity: $[3]_t A_{(2,2,1)}^{\text{alt}} + [2]_t A_{(2,1,2)}^{\text{alt}} + A_{(1,2,2)}^{\text{alt}} = m_{(2,2,1)} = P_{(2,2,1)}(x_1, x_2, x_3; t)$. (The Hall–Littlewood polynomial at this shape/rank collapses to a single monomial-symmetric term.)

Verdict: Conjectures $??$, $??$, $??$ all PASS.

3.4 Case 3: $\mu = (2, 2, 0, 0)$, $n = 4$ — the interesting case

$|B(\mu)| = 20$, $|\text{orbit}| = 6$, Bruhat lengths 0, 1, 2, 2, 3, 4. Atoms:

$$\begin{array}{ll} |A_{(2,2,0,0)}| = 1 & |A_{(2,0,2,0)}| = 2 \\ |A_{(2,0,0,2)}| = 3 & |A_{(0,2,2,0)}| = 3 \\ |A_{(0,2,0,2)}| = 5 & |A_{(0,0,2,2)}| = 6 \end{array} \quad \text{sum} = 20 = |B(\mu)|\checkmark$$

Weight $(1, 1, 1, 1)$: the multiplicity-2 ambiguous weight. Two SSYT: $[[1, 2], [3, 4]] \in A_{(0,2,0,2)}$ and $[[1, 3], [2, 4]] \in A_{(0,0,2,2)}$. These atoms are Bruhat-comparable: $(0, 2, 0, 2) \leq_B (0, 0, 2, 2)$ (lengths 3 and 4).

Coefficients of $x^{(1,1,1,1)}$ in the relevant A_γ^{alt} :

	$[x^{(1,1,1,1)}]$
$A_{(0,2,0,2)}^{\text{alt}}$	$(1 - t)^2$
$A_{(0,0,2,2)}^{\text{alt}}$	$(1 - t)^2$

This does not match Conjecture $??$'s literal reading. The naive attribution “coefficient $(1 - t)$ on diagonal, 0 above-diagonal” predicts $[x^{(1,1,1,1)}]A_{(0,2,0,2)}^{\text{alt}} = 1 - t$, whereas we observe $(1 - t)^2$.

But this *does* match Conjecture $??$: split

$$(1 - t)^2 = (1 - t) + t(t - 1) = (1 - t) + (-t)(1 - t).$$

The first summand $1 - t$ is the diagonal contribution (attributed to the SSYT $[[1, 2], [3, 4]] \in A_{(0,2,0,2)}$); the second summand $-t(1 - t) \in t \cdot \mathbb{Z}[t]$ is the above-diagonal contribution (attributed to $[[1, 3], [2, 4]] \in A_{(0,0,2,2)}$). Symmetrically for row $A_{(0,0,2,2)}^{\text{alt}}$: the $(1 - t)^2$ splits into diagonal $(1 - t)$ from $[[1, 3], [2, 4]]$ and *below-diagonal* shadow $-t(1 - t) = t(t - 1)$ from $[[1, 2], [3, 4]]$.

The $t = 0$ mod- t check. At $t = 0$: $A_\gamma^{\text{alt}}|_{t=0} = \sum_{T \in A_\gamma} x^{\text{wt}(T)}$ for all γ in the orbit, exactly recovering the classical Mason atom. Verified computationally for all 6 rows (see `analyze_ambiguous_weights.py`).

Global identity. $\sum c_\gamma A_\gamma^{\text{alt}} = P_{(2,2)}(x_1, x_2, x_3, x_4; t)$ with

$$c_{(2,2,0,0)} = (1 + t^2)(1 + t + t^2), \quad c_{(2,0,2,0)} = [2]_t[3]_t,$$

$$c_{(2,0,0,2)} = c_{(0,2,2,0)} = [3]_t, \quad c_{(0,2,0,2)} = [2]_t, \quad c_{(0,0,2,2)} = 1.$$

The coefficient $c_{(2,2,0,0)} = (1 + t^2)(1 + t + t^2) = [3]_t \cdot [2]_{t^2}$ is a t^2 -flavoured refinement — the first hint that the c_γ pattern for orbits with non-trivial stabiliser probes a two-parameter structure.

Verdict: Conjecture ?? PASSES; Conjecture ?? PASSES for all weights other than $\nu = (1, 1, 1, 1)$; the ambiguous-weight anomaly forces Conjecture ??, which PASSES.

4 Summary of verification

(μ, n)	$ B(\mu) $	orbit	Weak	Strong (unamb.)	Refined
$((2, 2, 0), 3)$	6	3	✓	✓	✓
$((2, 1, 0), 3)$	8	6	✓	✓	✓
$((2, 2, 1), 3)$	3	3	✓	✓	✓
$((2, 2, 0, 0), 4)$	20	6	✓	✓(13/14)	✓

The single “anomaly” in $((2, 2, 0, 0), 4)$ is the $(1 - t)^2$ on the multiplicity-2 weight, cleanly attributable under Conjecture ??.

Additionally, in all four cases the global identity $\sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}} = P_\mu(x; t)$ (Hall–Littlewood) is verified, with $c_\gamma \in \mathbb{Z}_{\geq 0}[t]$ following a Gaussian pattern that respects the orbit stabiliser structure.

5 Interpretation and next steps

5.1 Route δ^+ : promoted to load-bearing

Route δ^+ (the t -graded lift of Assaf–González’s local characterisation) is now supported by four verified cases spanning:

- all distinct parts ($\mu = (2, 1, 0)$),
- non-trivial stabiliser at $n = 3$ ($\mu = (2, 2, 0), (2, 2, 1)$),
- non-trivial stabiliser at $n = 4$ ($\mu = (2, 2, 0, 0)$).

Each case exhibits the block-lower-triangular structure of the coefficient matrix $M(\mu)_{\gamma, T}$, with the multiplicity-2 weight forcing a natural refinement.

Corollary 8 (Route δ^+ , conditional on Conjecture ?? uniformly). *If Conjecture ?? holds uniformly in μ , then the $V_{<\mu}$ consistency Lemma of the 07-23 partial proof closes as a structural statement about θ^{alt} -atoms: the coefficient matrix $M(\mu)$, together with the Gaussian coefficients c_γ , produces exactly $P_\mu(x; t)$ with residual $R = P_\mu - \sum c_\gamma A_\gamma^{\text{alt}} = 0$.*

This corollary is what motivates Route δ^+ as “cheapest to evaluate” (no DAHA needed — just polynomial verification at the mod- t level + t -divisibility of off-diagonal shadows).

5.2 Refined conjecture: uniform statement

Conjecture 9 (Uniform refined form). *For every partition μ (with n parts, zeros allowed), every $\gamma \in \mathfrak{S}_n \cdot \mu$, and every $T \in B(\mu)$, there is a canonical element $c_{\gamma,T}(t) \in \mathbb{Z}[t]$ such that:*

1. $A_{\gamma}^{\text{alt}} = \sum_{T \in B(\mu)} c_{\gamma,T}(t) \cdot x^{\text{wt}(T)}$;
2. $c_{\gamma,T_{\gamma}}(t) = 1$ for the extremal $T_{\gamma} \in A_{\gamma}$;
3. $c_{\gamma,T}(t) \equiv \mathbf{1}_{T \in A_{\gamma}} \pmod{t}$;
4. $c_{\gamma,T}(t) \in t \cdot \mathbb{Z}[t]$ for $T \notin A_{\gamma}$;
5. For $T \in A_{\gamma}$ non-extremal: $c_{\gamma,T}(t) - (1 - t) \in t \cdot \mathbb{Z}[t]$.

Equivalently: the coefficient matrix $M(\mu)$ has classical Mason atom support structure at $t = 0$, with all deviations from $\pmod{t}$ Bruhat-block-lower-triangular in the sense of PROVE.md's original conjecture.

Verified: $((2, 2, 0), 3)$, $((2, 1, 0), 3)$, $((2, 2, 1), 3)$, $((2, 2, 0, 0), 4)$.

5.3 Why the multiplicity-2 case is not a bug

The $(1 - t)^2$ coefficient on $x^{(1,1,1,1)}$ has a plausible structural explanation: it counts the two independent “paths” from the highest weight to the two SSYT of weight $(1, 1, 1, 1)$ in the Kashiwara crystal, each contributing a $(1 - t)$ factor. At $t = 0$, each SSYT contributes 1; multiplying gives $1 = 1 \cdot 1$, matching the classical count. At generic t , the interaction between the paths produces the $(1 - t)^2$ factor.

This suggests a *combinatorial refinement* at the crystal-graph level: the coefficient $c_{\gamma,T}$ might be a sum over paths from T_{γ} (extremal in A_{γ}) to T in a suitably-oriented Kashiwara subgraph, weighted by t -factors according to the path structure. This is speculative and left as a next-step direction.

5.4 Priority for next PROVE cycle

1. **Verify at $\mu = (3, 2, 1)$, $n = 3$** — all distinct parts, larger partition, tests whether c_{γ} pattern generalises beyond $|\mu| \leq 4$.
2. **Verify at $\mu = (2, 2, 2, 0)$, $n = 4$** — non-trivial stabiliser at $n = 4$, but different from $(2, 2, 0, 0)$; tests whether $c_{\gamma} = [3]_t[2]_{t^2}$ -style two-parameter structure persists.
3. **Verify at $\mu = (3, 1, 0)$, $n = 3$** — gap > 1 parts, to test whether the c_{γ} structure is orbit-shape-invariant (as PROVE.md hints for the atoms-side pattern).
4. **Formalise Conjecture ?? at all μ with $|\mu| \leq 6$** — three probes above, plus $(3, 1, 1, 0)$ and $(2, 1, 1, 0)$ at $n = 4$.
5. **Proof sketch attempt** of the mod- t part of Conjecture ??: at $t = 0$, this is the classical L.-S. 1990 / Mason 2006 statement that Demazure atoms partition the Schur function.

5.5 Byproduct: Dim Lemma stability

The Dim Lemma of the 07-24 partial proof (dim of S_n -symmetric functions in V_μ^{alt} is ≤ 1) uses only the mod- t block-triangular structure — Conjecture ?? part (3). Since this part is verified in all four cases and is a known classical fact (L.–S. 1990), the Dim Lemma holds *unconditionally* and can be shipped as a 4-page standalone note independent of the full $V_{<\mu}$ resolution.

6 Computational verification

Code lives at:

- `~/projects/probes/2026-07-24-t-graded-A-G-verification/verify_block_triangular.py` (main verification script; runs all four cases in < 10 seconds).
- `.../verify_global_identity.py` (solves for c_γ and verifies $\sum c_\gamma A_\gamma^{\text{alt}} = P_\mu$; uses Macdonald III.(2.6) formula for P_μ).
- `.../analyze_ambiguous_weights.py` (unpacks the multiplicity-2 weight case at $n = 4$; verifies the mod- t exact match to classical Mason atoms).

All checks pass.

References

- [1] S. Assaf and N. González. *A local characterization of unions of Demazure crystals*. arXiv:2512.19814, December 2025.
- [2] S. Mason. *A decomposition of Schur functions and an analogue of the Robinson–Schensted–Knuth algorithm*. Sémin. Lothar. Combin. 57, Article B57e, 2006/07.
- [3] A. Lascoux and M.-P. Schützenberger. *Keys and standard bases*. IMA Vol. Math. Appl. 19, 125–144, 1990.
- [4] I. G. Macdonald. *Symmetric Functions and Hall Polynomials*, 2nd ed. Chapter III. Oxford, 1995.
- [5] Clio. *Atom decomposition of P_μ via $KI + \text{Cherednik–Ram}$ — partial proof*. `~/projects/proofs/2026-07-23-atoms-decomposition-P-mu.tex`, 2026.
- [6] Clio. *P - δ probe: verified at $\mu = (2, 2, 0), n = 3, t = 0$* . `~/projects/proofs/2026-07-24-p-delta-probe-Vmu-at-tzero.pdf`, 2026.
- [7] Clio. *Atom decomposition of P_μ via $KI + CR$ — Part II: Reduction Theorem*. `~/projects/proofs/2026-07-24-atoms-decomposition-P-mu-part-II.tex`, 2026.